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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION – March 2026

Fourth Semester – Discipline Specific Core Course (DSC)

MG4DSCZGY201 - BIOLOGICAL CHEMISTRY

Time: 1.5 hrs **Maximum marks: 50**

Part A – Multiple Choice Questions

Answer all questions. Each question carries 1 mark.

1. A complete, catalytically active enzyme consisting of apoenzyme and cofactor is known as: **[K] / [CO2]**
    - **Answer: c). Holoenzyme**
2. Fructose-1,6-bisphosphatase converts fructose-1,6-bisphosphate to: **[K] / [CO4]**
    - **Answer: a). Fructose-6-phosphate**
3. Carbohydrates containing a ketone group are called: **[K] / [CO1]**
    - **Answer: d). Ketoses**
4. Anaerobic respiration in yeast produces: **[K] / [CO4]**
    - Answer: b). Ethanol and CO2
5. Which of the following amino acids is purely ketogenic? **[K] / [CO1]**
    - **Answer: b). Leucine**
6. Cholesterol is delivered to peripheral tissues by: **[K] / [CO4]**
    - **Answer: a). LDL**
7. Which of the following is a function of phospholipids? **[K] / [CO1]**
    - **Answer: c). Formation of biological membranes**
8. Which molecule acts as an allosteric activator of PFK-1? **[K] / [CO2]**
    - **Answer: c). AMP**
9. Starch is digested by salivary and pancreatic amylase into: **[K] / [CO1]**
    - **Answer: b). Maltose**
10. Bee wax is an ester of: **[K] / [CO1]**
    - **Answer: c). Palmitic acid and a long chain alcohol**
11. During glycogenolysis, glycogen phosphorylase cleaves glycogen to produce: **[K] / [CO4]**
    - **Answer: c). Glucose-1-phosphate**
12. NADH stores energy in the form of: **[K] / [CO3]**

- Answer: b). Electrons

Part B – Short Answer Questions

Answer any 6 questions. Each carries 3 marks.

13. Mention the biological functions of RNA. [U] / [CO1]

RNA definition & full form- 1 mark; any two functions- 2 marks

Key Points: mRNA carries genetic code for protein synthesis; tRNA translates codons into amino acids; rRNA forms the core of ribosome structure.

14. List the major risk factors for atherosclerosis. [U] / [CO4]

Any 3 factors- 1 mark each

Key Points: High LDL cholesterol, hypertension (high blood pressure), tobacco smoking, diabetes mellitus, and sedentary lifestyle.

15. Explain the activation energy in enzymology. [U] / [CO2]

Explaining activation energy & transition state- 2marks; Graphical representation – 1 mark

Key Points: The energy barrier that must be overcome for a reaction to proceed; enzymes function by lowering this energy to increase reaction rates.

16. What is oxidative deamination? Name the enzyme involved. [U] / [CO4]

Defenition – 2; Enzyme- 1

Key Points: Removal of an amino group from an amino acid as ammonia while oxidizing the carbon skeleton; primary enzyme is **Glutamate dehydrogenase**.

17. Explain the importance of NAD+ in glycolysis and fermentation. [U] / [CO4]

Liberal valuation

Key Points: NAD+ is an essential oxidizing agent; in fermentation, it is regenerated from NADH to allow glycolysis to continue producing ATP in the absence of oxygen.

18. Explain DNA enzymes (Deoxyribozymes). [U] / [CO2]

Any 3 points

Key Points: Non- protein enzymes, Single-stranded DNA molecules capable of catalyzing specific chemical reactions, such as RNA cleavage or ligation, artificially created.

19. Describe exergonic reactions. [U] / [CO3]

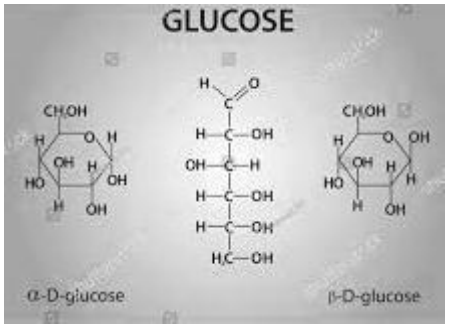
Definition- 1.5 marks; Explain with an example (hydrolysis of ATP)- 1.5 marks

Key Points: Reactions where the change in free energy is negative (Delta G < 0); they release energy and occur spontaneously.

20. Describe the structure of glucose. [U] / [CO1]

Description with open chain or ring structure

Key Points: An aldohexose (C6H12O6) with six carbons and an aldehyde group; typically exists in a cyclic hemiacetal form (pyranose ring).



Part C – Short Essay Questions

Answer any 4 questions. Each carries 5 marks.

21. Discuss how enzymes increase the rate of a chemical reaction. [U] / [CO2]

Enzyme definition- 1 mark; Activation energy and transition state- 3 marks

Mention Enzyme substrate complex- 1 mark

Mechanism: Enzymes act as biological catalysts by lowering the **activation energy** required for a reaction to proceed. They provide an alternative reaction pathway where the transition state is stabilized.

Binding: The formation of an **Enzyme-Substrate (ES) complex** at the active site positions substrates in the optimal orientation for the reaction to occur.

22. Write an account on molecular chaperones and their biological importance. [U] / [CO1]

Definition and any 4 points- 4 marks; Examples- 1marks

Liberal valuation

Chaperones are specialized proteins that assist in the correct folding of newly synthesized polypeptide chains.

They prevent "misfolding" or aggregation of proteins, which is critical for maintaining cellular health and preventing diseases like Alzheimer's or Parkinson's.

23. Describe beta-oxidation of saturated fatty acids. [U] / [CO4]

Definition- 1 mark; six steps- 3 marks; example (palmitic acid or any other)- 1 mark

Process: This is the mitochondrial pathway for breaking down fatty acids into **Acetyl-CoA** units.

Stages: It involves a six-step cycle**: 1. Activation of FA, 2. Carnitine shuttle, 3. Desaturation, 4. Hydration 5. Oxidation (NAD-linked), and 6. Thiolysis**. Each cycle shortens the fatty acid chain by two carbons.

24. Summarize the importance of polysaccharides and stereoisomers. [U] / [CO1]

Liberal valuation

Definition of polysaccharides and sterioisomers: 2 marks

Examples: 2 marks

Importances: 1 mark

Polysaccharides: Serve as structural components (cellulose in plants) or energy storage (starch in plants, glycogen in animals). **Stereoisomers:** The spatial arrangement of atoms (like D- vs L-glucose) is vital because biological systems and enzymes are **stereospecific**; they usually only recognize one specific isomer.

25. Explain uncompetitive inhibition and describe how it affects enzyme-substrate complex formation. [U] / [CO2]

Liberal valuation

Definition- 2 marks; an example- 2 marks; effect on Vmax and Km- 1 mark

Definition: In this type of inhibition, the inhibitor binds **only** to the Enzyme-Substrate (ES) complex, not the free enzyme.

Effect: It effectively "locks" the substrate in the active site, preventing the reaction from completing. This results in a decrease in both the maximum velocity Vmax and the Michaelis constant (Km)

26. Describe the overall energy yield during the oxidation of glucose. [U] / [CO4]

Energy yield during glycolysis- 1 mark; pyruvate oxidation- 1 mark

Kreb cycle and ETC- 2 marks. Correct calculation mentioning conversion of FADH2 and NADH to ATP- 1mark.

Glycolysis: Yields 2 ATP and 2 NADH.

Citric Acid Cycle (TCA): Following the link reaction, the TCA cycle produces GTP (ATP equivalent), NADH, and FADH2

Total Yield: Through the Electron Transport Chain (oxidative phosphorylation), the complete aerobic oxidation of one glucose molecule typically yields approximately **30 to 32 ATP**.